

REVIEW

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Comprehensive review of methods for leaf disease identification

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Abstract

Automatic image classification for plant leaf disease identification (LDI) is an important task in computer vision, food processing, robotics and precision agriculture. Humans classify diseases that occur in plants by careful examination of leaves via visual inspection and knowledge of diseases that can occur to various plants. Similarly, digital images of leaves acquired through devices or from existing datasets are examined via computer vision techniques and classified with trained knowledge using various artificial intelligence approaches. This review discusses several methods explored by various researchers and visualizes their results in terms of the accuracies achieved and also the limitations of each method related to leaf disease identification. The objective of this article is to present a comprehensive review of recent research works by briefly describing the nature, size of data, No of plants and diseases covered, steps involved in the classification approaches, performance and limitation. Thus this article gives overview of recent machine learning (ML), deep learning (DL) and other approaches in LDI. The correctness of LDI classification is presented to the readers by mentioning the accuracy as found from the research articles. Apart from this other efficiency considerations are presented as when needed for describing the research work. This would give a quick summary of review of latest LDI research works by providing black box presentation to the approaches without elaborating the detail steps of the chosen approach.

Keywords Image classification, Leaf disease identification, Machine learning, Deep learning, Unsupervised approaches, Self-supervised approaches

1 Introduction

1.1 Background

The automation of LDI is the process of developing a system that detects leaf diseases automatically with no human intervention. In the context of technology and agriculture, automation involves the use of software, algorithms, and sophisticated machinery to automate complex tasks. The LDI system accepts leaf images as input and returns a label to that image. Several steps are involved in automation which comprises pre-processing, feature extraction, classification and evaluation. Researchers have explored copious options in all these steps and have recorded their results. The methods that achieve good performance in terms of accuracy are identified and the other methods can be



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avoided because they are not suitable for LDI. Along with these studies the limitations of a method as specified by researchers are expository reviewed and analyzed and areas where research work has not been extended are found.

Research articles in the domain of leaf disease identification published after 2018 were collected and systematically reviewed by analyzing the dataset they used and their methods including pre-processing, feature extraction, optimization and performance achieved in terms of accuracy as described in [1, 2]. Signs and symptoms of diseases and different neural Network architectures are briefly shown in the diagrams in [3]. A review of ML and DL classification techniques was performed in a comparative study by [4]. Other reviews [5–13] reviewed leaf disease identification methods. Approaches for disease detection in potato plants were reviewed in [14].

1.2 Review methods

The relevant literatures from 2018 were obtained by searching with the term ‘leaf disease identification’ in Google, IEEE and Scopus.

Peer reviewed and journal articles are collected for the study. From the study articles were chosen on the basis of two points.

1. Articles describing a new process where input should be leaf images and output should be leaf images with labels and the articles that can be downloaded or viewed fully through our institution subscription.
2. Articles from references and citations in those articles and which describes a process as in point 1.

The method for searching and collecting existing articles was depicted in Fig. 1. The percentage of articles from the total 120 that are present in the databases like SCOPUS, WoS and IEEE explore are shown in Fig. 2.

The collected research articles were grouped by the classification method used for LDI for further study. In a study [15] analyzed various feature extraction techniques, summarized common diseases, their symptoms and the pathogens responsible for various plant categories and grouped various classification techniques into supervised, unsupervised

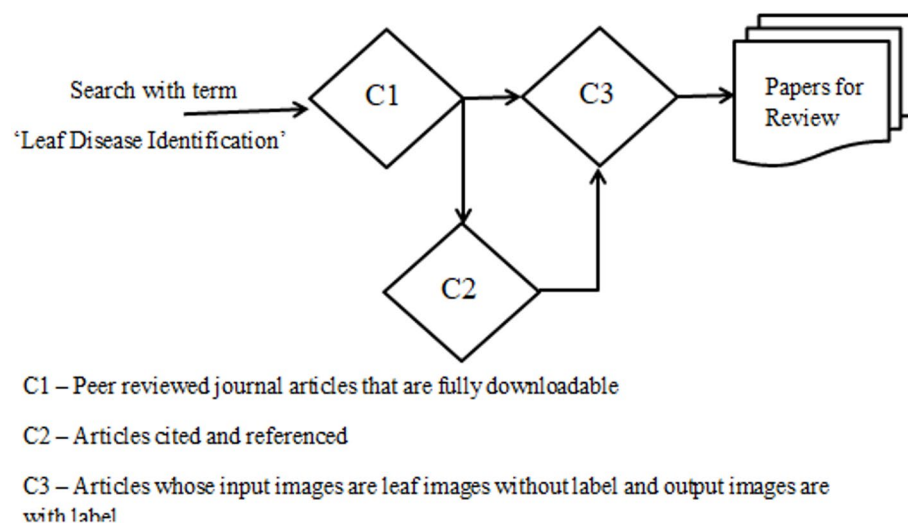


Fig. 1 Review paper collection method

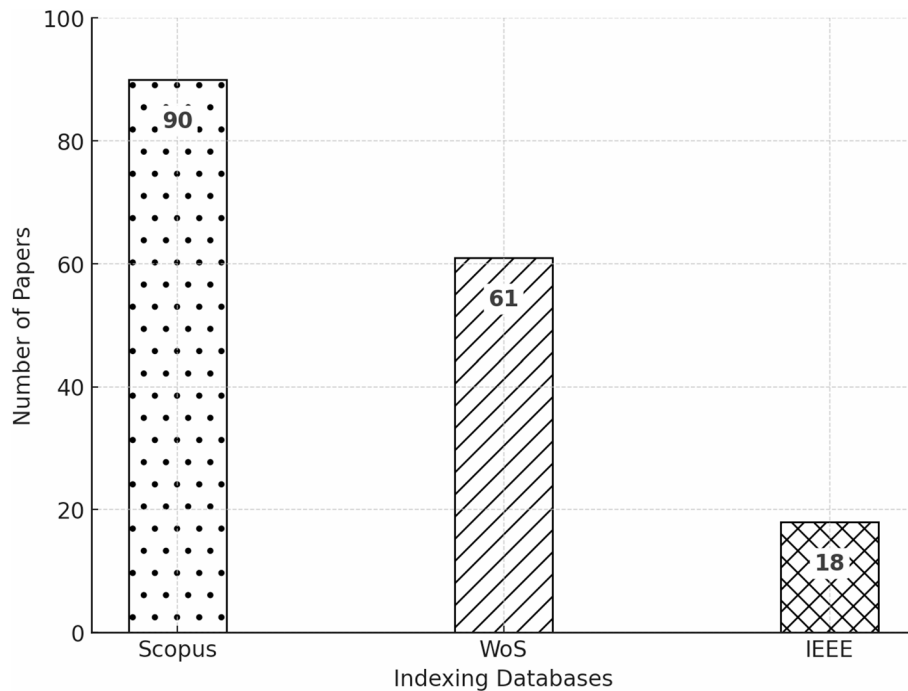


Fig. 2 Number of papers indexed in different databases

and others. One more approach self-supervised was added to this comprehensive review. Each group comprises quite a few classification methods. Within each classification method they were arranged in chronological order based on year of publishing. Multiple papers on the same year are arranged by the increasing number of plants or diseases or both. Since there are more papers using CNN and SVM, they were further sub grouped.

1.3 Review highlights

Table 1 lists the highest accuracies of each classification method for the three types of dataset from among the reviewed papers. In most of the papers accuracy was presented in percentage. In some papers where the accuracy was not in percentage it was converted to percentage. The three types of dataset are laboratory, real and other. The hyper-spectral, multispectral and Infrared (IR) images contribute to the other images. The second column of Table 1 shows the number of papers (NOP) where particular classification methods are implemented. Some researchers have implemented different classifiers in the same experimental setup and have compared the results. So the total NOP was 155 whereas the total number of references was 120. Among the total 120 papers 23 survey and review articles and 98 implementation papers were reviewed.

The NOP where the CNN, its pre-trained models or a specialized CNN contribute to 79 research works whereas all the other classification methods together are found only in 76 research works. Even though this statistical analysis is only from the articles we have collected for our review it strongly projects the CNN as the best classifier for leaf disease identification on the basis of performance. Compared with other classification methods it is also robust to noise in real images. It has to be trained with a large dataset for achieving this accuracy and it also requires a high-end processor such as a GPU when the chosen dataset includes multiple plants and multiple diseases. Nevertheless CNNs have not been used for images other than the visual spectrum.

Table 1 Related literature review – highest accuracies in %

Classification Approaches	NOP 155	Type of images of dataset		
		Laboratory	Real	Others
Unsupervised Approaches	10			
Discriminant Analysis	6	93.5 [16]	–	95 to 100 [17]
Variational AutoEncoder (VAE)	2	98.27 [18]	–	–
Iterative Self Organizing Maps (ISO)	1	–	–	35 to 63 [19]
MRNN	1	Accuracy im- proved [20]	–	
Supervised Approaches	134			
Naïve Bayes (NB)	2	84.69 [21]	61–90% [22]	
K nearest neighbour (KNN)	7	93.8 [16]	–	66 [23]
Decision Tree (DT)	3	94.5 [16]	–	–
Random Forest (RF)	5	100 [24]	–	90.69 [25]
Region Proposal Network (RPN)	1		83.57 [26]	–
Back Propagation Neural Network (BPNN)	2	97 [27]	89.12 [28]	
Support Vector Machine (SVM) (20)	10	99 [29]	84.44 [28]	90.67 [25]
Kernel Functions	6	98.71 [30]	–	99.98 [31]
Clustering techniques for Segmentation	4	98.71 [30]	84.3 [32]	–
Feed Forward Neural Network (FFNN)	1		98.1 [33]	
Multi_Layer Perceptron (MLP)	4	99.1 [34]		99–100 [35]
Radial Basis Functional Neural Network (RBFNN)	2	95 [36]	94.20 [37]	–
Convolutional Neural Network (CNN) (79)	5	99.74 [38]	98 [39]	–
Pre-trained models				
AlexNet and its variants	7	99.44 [40]	97.63 [41]	–
VGG and its Variations	11	99.48 [40]	90 [42]	–
ResNet and its Variants	11	99.62 [43]	99.78 [44]	87 [45]
GoogleNet and its Variants	8	99 [46]	98.9 [47]	–
MobileNet and its Variants	6	99.5 [48]	–	–
DenseNet and its variants	6	99.74 [49]	–	–
Inception and its variants	6	99.60 [50]	98.43 [41]	–
Xception and its variants	2	100 [46]	–	–
EfficientNet and its Variations	6	100 [51]	96.02 [52]	–
Other CNN	8	99.78 [53]	98.7 [54]	–
Multi-Layer CNN	3	96.46 [55]	97.13 [37]	–
Attention Network	4	99.43 [56]	89 [57]	–
Capsule Neural Networks	1	–	–	–
Self-Supervised Approaches	3			
Location Feedback Control Network (LFC-Net)	1	99.7 [58]	–	–
Vision Transformer (ViT)	2	100 [59]	96.85 [60]	–
Other Approaches	8			
Statistical	1	91.88 [61]	–	–
Fuzzy	3	98 [62]	–	–
Hybrid	4	98.38 [63]	–	97.62 [64]

The complexity of these different dataset varies. Laboratory images are captured under a controlled environment. They include some or all of the following factors: fixed background, standard lighting source providing uniform illumination effects, common image acquisition devices, distance from the leaf to the device, and the angle of the device. The image will contain a single leaf in most cases. There are many standard datasets available for this type of images which contain huge collections of leaf images of numerous plants and diseases.

In real images, the capturing person may be a farmer or a common man who does not have knowledge about the camera or lighting effects. The images would contain background clutter, lighting variations and other distortions such as mud, dew or pests. This will increase the complexity of the classification process. Segmentation should also deal with noise removal in these images. Researchers have used collected images or images from private reference for real images. They are usually in hundreds. Some standard dataset also contains small number of images captured in field. But they are captured by focussing a single leaf of the plant which holds reduced complexity.

The hyperspectral images are obtained using a hyperspectral sensing system that captures light radiations other than the visible spectrum (VIBGYOR—Violet, Indigo, Blue, Green, Yellow, Orange & Red) and they also include infrared(IR) or other spectral ranges. It would be helpful in detecting diseases that are not visible to the human eye, especially during the early stage of the disease. The number of images available in this spectrum is very low. There is no standard dataset available for these images. Researchers used collected data of images of leaves inoculated with disease pathogens.

1.4 Organization of paper

The review of related literature as per the groupings in Sect. 1.2 was described in Sect. 2 Literature Review followed by Sect. 3. Limitations and Discussion. Section 2 Related Literature Review is elaborated with subSects. 2.1 unsupervised approaches, 2.2 supervised approaches. 2.3 self-supervised approaches and 2.4 other approaches. Each of these approaches will have classification methods as subsection headings. As CNN and SVM classification was carried out in many studies they were further divided into subsections. SVM subsections were a. Kernel Functions and b. Clustering techniques for segmentation. The subsections of CNN were given in Tables 2, 3 4 and 5.

These sections give raw data of existing research works rather than a cohesive literature review. But this would be helpful for new researchers to integrate narrative comparisons and draw generalized insights.

2 Literature review

2.1 Unsupervised approaches

Leaf diseases are classified without prior training or target labels by detecting anomalies or by building generative models for typical inputs. Discriminant analysis, the variational autoencoder (VAE), iterative self-organizing (ISO) and the modified recurrent neural network (MRNN) are unsupervised approaches that have been used in various studies for the classification of leaf disease.

Unsupervised approaches can learn patterns and features directly from unlabeled leaf images and so the training phase is not done with labeled data initially. After categorization, the category is labeled. Hence the training time is minimized than the supervised classification. When it comes to leaf images in a complex real environment, segmentation is necessary before categorization for improved accuracy. Unsupervised approaches can be effective for leaf disease identification, especially when labeled data are scarce and acquiring large labeled datasets is challenging for some plants or diseases.

Table 2 Pretrained models with one or more diseases on a single plant

Citation	Proposed Work and dataset	CNN Model and Accuracy
[47]	The improved GoogLeNet and Cifar10 models based on deep learning classified 8 Maize leaf diseases identified from 500 images collected from PV dataset and Google websites	GoogLeNet 98.9% Cifar10 98.8%
[65]	k means ResNet used for Maize Kernel Classification while Vision System was established for capturing maize kernel	ResNet 98.2%
[41]	Six Neural Network Architecture used for 4 diseases Six DL, Machine Learning and Manual analyzed for Grapevine Yellow disease. Grapevine Images photographed as well as from PV dataset System ACC DL 0.9206 (0.0039), ML 0.2679 (0.0089), Manual 0.7569 (0.0544)	AlexNet 0.9763 GoogLeNet 0.9636 Inception v3 0.9843 ResNet-50 0.9918 ResNet-101 0.9933 SqueezeNet 0.9377
[42]	VGG 16 for images Segmented with SVM and Augmented with Conditional Deep Convolutional GAN. Tea Leaf 3 diseases from images captured using Digital Camera in hand and UAV and database images	C-DCGAN VGG16—90%
[66]	Hybrid Features LBP and color texture-Deep CNN VGG16 5569 Tea Leaf images captured indoors. Classified for 4 diseases	LBP VGG16 96.20%
[67]	3ML (SVM, RF, Stochastic Gradient Descent (SGD)) and 3DL (Inception-v3, VGG-16, VGG-19) methods—Image embedding is used to provide the format that can be handled by ML—SqueezeNet(SN) is utilized. Images are given as input and produce the vector of numbers that could be used for further processing using ML algorithms.—works faster and an internet connection is not mandated. Citrus plant disease detection	VGG-19 87.4% Inception-v3 89% VGG-16 89.5%
[43]	Transfer Learning and Baseline Learning Methods were adopted for Learning. SGDM, ADAM and RMS Propagation were used for training. From 38 classes of Potato leaf disease from PV dataset 2 diseases were classified	ResNet99.62% VGG16 98.88% Inception V3 98.00% VGG19 97.40% NasnetV2 96.00% MobileNetV2 94.56%
[68]	Common bean 3 diseases were classified by retraining the pre-trained models AlexNet, GoogleNet, VGG16 with 1766 collected images	Proposed Model 98%
[69]	Evaluated four deep neural network models trained with different parameters for fine-tuning (FT) and transfer learning improving the identification accuracy. soybean leaf diseases applied to images taken straight from a small and cheap unmanned aerial vehicle (UAV)	Inception-V3 FT75% 99.04% ResNet-50 FT75% 99.02% VGG 19 FT100% 99.02% Xception FT100% 98.56%
[70]	Grape Leaf 4 diseases classified with the LeafGAN generated dataset and without LeafGAN generated dataset. with eight common image classification models AlexNet, VGG-11, ResNet-34, DenseNet-121, Xception, ResNext-50, SEResNet-50, EfficientNet-b0 were implemented to train and test these datasets	AlexNet 94.60 VGGNet 93.75 ResNet 96.32 DenseNet50 98.13 Xception 98.10 ResNext 98.33 SEResNet 97.60 EfficientNet 98.25
[71]	1500 Tomato leaf images of 5 different classes from PV dataset—AlexNet, GoogLeNet, ResNet and VGG16—GoogLeNet had the highest Accuracy	DCGAN AlexNet 89.67% DCGAN GoogLeNet 94.33% DCGAN ResNet 83% DCGAN VGG16 79.3%
[72]	Based on DenseNet-121 deep convolution network, three methods of regression, multi-label classification and focus loss function were proposed AI challenger dataset 2462 Apple leaf images 6 diseases	DenseNet121 Average Accuracy approximately 93%
[50]	Four CNN models were used for performing 16 experiments and measure classification accuracy in each method. Eight methods with Feature Extraction on Model Laboratory dataset and field dataset. Other eight methods with parameter-tuning on Model laboratory dataset and Field dataset. tomato leaf 5 diseases on laboratory-based dataset and self-collected data from the field	VGG-16 88.10 76.20 VGG-19 90.00 79.60 ResNet 91.70 81.70 Inception V3 93.40 85.00 VGG-16 98.50 84.10 VGG-19 98.30 86.30 ResNet 99.40 91.30 Inception V3 99.60 93.70

Table 2 (continued)

Citation	Proposed Work and dataset	CNN Model and Accuracy
[73]	An apple disease classification model based on multi-scale feature fusion is proposed in this paper. Conventional residual network (ResNet) was improved. Stochastic Gradient Descent (SGD) was used to optimize the model. Apple Dataset from Kaggle and Baidu AI studio were collected in a natural environment for 5 kinds of Diseases	Improved ResNet highest accuracy of 94.23% in original dataset and 94.99% in preprocessed dataset
[46]	4 CNN models for Augmented tomato leaf images. 109,290 images of 1 plant for 9 diseases from PV dataset	MobileNetV2 0.75 0 NasNetMobile 0.84 Xception 1.00 MobileNetV3 0.98 AlexNet 0.98 GoogLeNet 0.99 ResNet18 0.99
[74]	Tomato plant Diseases from PV dataset—EfficientNet Models—EfficientNet-B0, EfficientNet-B4, and EfficientNet-B7—used for binary classifier- Six class classifier—10 class classifier	Two-class 99.95% Six-class 99.12% Ten-class 99.89%
[75]	Proposed model is an ensemble of pre-trained DenseNet121, EfficientNetB7, and EfficientNetNoisyStudent. Dataset from Cornell University -3642/3651 images of Apple leaf – 3 Diseases	DenseNet 92.60 GoogleNet 95.30 EfficientNetB7 95.62 ResNet20 93.70 VggNet-16 94.00 Proposed Model 96.25
[76]	Proposes DenseNet201 experiments four CNN models, VGG16, Inception V3, ResNet152V2 and DenseNet201. Tomato plant 9 diseases—22,930 images from PV dataset	VGG16 94.47% Inception V3 92.58% Resnet152V2 96.03% DenseNet201 96% DenseNet201 and Transfer Learning 97.97%
[77]	ResNet and MobileNet are used, MobileNet outperforms. Pomegranate-Mendeley Database	MobileNet 97.32%
[78]	A pre-trained Efficient DenseNet model has been employed utilizing an extra transition layer in DenseNet-201 to classify the potato leaf diseases efficiently. Potato plant 4 diseases from PV dataset and gathered images	DenseNet-201 96.03 Efficient DenseNet 97.2
[79]	A multi-modal feature fusion network, combining a RGB image network and hyperspectral band extraction network, was proposed four categories (HLB, suspected HLB, Zn-deficient, and healthy) Captured images of RGB Camera and hyperspectral camera for Citrus Plant	ResNet50 + hyperspectral feature extraction network 96 VGG16 + hyperspectral feature extraction network 95.1 ResNeXt101 + hyperspectral feature extraction network 97.89
[45]	hyperspectral images of strawberry leaves that were inoculated with graymold fungus to cause disease were taken for classification into healthy and infected with the proposed 3D CNN model that can use the input structure of $16 \times 16 \times 150$ hyperspectral data without modification, and the results showed a training accuracy of 0.87 and a validation accuracy of 0.83. In addition, a 3D CNN that used extended input ultimately had a training accuracy of 0.89 and a validation accuracy of 0.84	ResNet 50 87%
[80]	Citrus plant Fruit disease and leaf disease using deep learning and image processing. Citrus plant Fruit disease and leaf disease for 6 diseases from citrus disease image gallery dataset and the combined dataset	Without SGDM Alexnet 92.2% VGG19 91.7% With SGDM Alexnet 94.3% VGG19 92.9%
[48]	Single stream convolutional neural network architecture is proposed for recognizing citrus fruit diseases. One Plant fruits, leaves, and hybrid dataset. 607 images in Leaves Dataset were considered	MobileNet-V2 CNN Fruits 99.4, Leaves 99.5 and Hybrid 99.7%

Table 3 Pre-trained models applied to multiple plants with multiple diseases

Citation	Proposed Work and Dataset	CNN Model and Accuracy
[40]	5 CNN models were developed, trained and tested, 87,848 images of 25 plants, 58 Diseases from PVdataset	Original & Preprocessed Image AlexNet 99.06% 98.64% AlexNetOWTBn 99.44% 99.07% GoogLeNet 97.27% 97.06% Overfeat 98.96% 98.26% VGG 99.48% 98.87%
[49]	VGG16, ResNet50 and DenseNet121 used for classification. Accuracy for the dataset expanded with DoubleGAN is higher than that for the original dataset and dataset expanded by flipping and translation.—31,361 leaf images 5 different species 10 classes from PV	Flipping and translation expansion VGG16 99.36% ResNet50 98.93% DenseNet121 99.74% DoubleGAN expansion VGG16 98.58% ResNet50 99.14% DenseNet121 99.70%
[81]	GoogLeNet—1567 images representing 79 diseases affecting 14 plant species. Approximately 60% of the images were captured under controlled conditions, and 40% under real field conditions	GoogLeNetMildly diseased 31% Moderately diseased 87% Severely diseased 94%
[51]	EfficientNet is improved using compound scaling ensembling on feature vectors. EfficientNet family consists of 8 progressively improved versions (b0-b7)—Pretrained models from ImageNet were used.—Augmented version of PV dataset 14 Plants 39 classes	End-to-end 99.9729% 99.9904% original Augmented Fine-tuning 99.9856% 99.9919% Minimal ensemble 100.000% 100.000%
[82]	Deep convolutional-neural-network (CNN) models implemented EfficientNet B0 have high accuracy –PV dataset 14 different plant species 38 Diseases	InceptionV3 98.42 InceptionResNetV2 99.11 MobileNetV2 97.02 EfficientNetB0 99.56
[52]	a novel deep ensemble neural network (D-ENN) Inceptionv3, MobileNetv2, DenseNet121, and EfficientNetB0 with C-GAN for Vegetable leaves 11 diseases. Custom field images with wide variations in illumination and weather conditions	Deep Ensemble Neural Network D-ENN 96.02%

2.1.1 Discriminant analysis

The linear discriminant analysis (LDA) in [16] achieved 93.5% accuracy for classifying six diseases of citrus plants as a comparative model. However, the accuracy decreased to 85.3% for the combined dataset of fruits and leaves and 83.2% for the locally recorded dataset consisting of RGB images. The classification method was not tested on real time images, hyperspectral images or multispectral images.

Quadratic discriminant analysis (QDA) yielded classification accuracies ranging from 75–99% for grapevine leafroll disease (GLD) from field data comprising visible and infrared images from two years and six seasons and the authors planned to focus on early-stage detection in the future [22].

Research involving hyperspectral images of tomato leaves was performed via stepwise discriminant analysis (STDA) without any pre-processing or feature extraction in two works. Four categories healthy, asymptomatic, early and late disease development stages were classified by testing with hyperspectral wavebands and different vegetative indices [35], while 5 out of 35 vegetative indices were identified as the best and had accuracies ranging from 80 to 88. Another work [17] classified leaves into 4 categories (healthy, BS, TS and TYLC) by using two vegetative indices the renormalized difference vegetation index (RDVI) and the modified triangular vegetation index (MTVI) and had accuracies of various disease stages ranged from 95 to 100.

Table 4 Other convolutional neural networks

Citation	Proposed Work and Dataset	CNN Model and Accuracy
[83]	A novel neural network is designed based on backbone Alexnet architecture, named DMS-Robust Alexnet for maize leaf of 7 classes, involving a total of 12,227 pieces collected in the Natural environment. Dilated convolution and multi-scale—convolution are combined to improve the capability of feature extraction	SEG-KNN 80.12 GA-SVM 82.71 Simple-CNN 87.46 ResNet50 95.47 GoogleNet 94.06 DMS-Robust Alexnet 98.62%
[84]	MultiModel_VGR is proposed for automatic detection and severity assessment of pepper bacterial spot disease from PV dataset, based on VGGNet, GoogLeNet and ResNet	MultiModel_VGR 95.34%
[85]	A lightweight, fast, and accurate deep metric learning-based architecture is proposed. It is a patch-based classification network that comprises an embedding module, a cluster prototype module, and a simple neural network classifier. 609 citrus images for four classes were trained and tested with tea leaf images from public dataset	Proposed Model 112 95.04% 98.33% Proposed Model 128 95.04% 98.16%
[86]	DenseNet pre-trained on ImageNet and Inception module (DENSE-INCEP)—Rice plant 3 diseases on Public Dataset UCI repository and with Rice plant 12 diseases on real time own dataset	Average Accuracy Public Dataset 3 diseases 94.07% Field Dataset 12 diseases 98.63%
[87]	The VGGNet pre-trained on ImageNet and Inception modules are selected in our approach.—Trained on on-field images and tested on PV dataset	INC-VGGNet—92% Rice dataset INC-VGGNet—80% Maize dataset
[88]	Valid lightweight network architecture, namely, MobInc-Net, to perform crop disease recognition and detection. -Crop Images Laboratory and on field	accuracy of 99.21% on the public dataset and 97.89% on the local dataset
[8]	proposed KijaniNet model for semantic segmentation and Classification done with CNN—Mobile captured images as well as PDDb dataset	KijaniNet 94.99%
[89]	Crop pest Image Recognition methods using CNN and Transfer Learning Methods. Each image Augmented and Deaveraged. – pre-trained on ImageNet large image dataset and verified on Image Database for Agricultural Diseases and Pests (IDADP). 15 plants	AlexNet, VGG-16, and ResNet-50 achieving accuracy of around 98%
[54]	automatically evolving CNN architecture using big bang–big crunch (BB–BC) algorithm—Rice Plant 4 diseases 5932 on-field rice leaf images -	CNN using BB-BC 98.7%
[44]	Image divided into patches-Clustered-Weakly supervised CNN—ResNet—271 plant disease categories and 220,592 images	ResNet152 99.78%
[90]	A texture-color dual-branch multiscale residual shrinkage network (TC-MRSN) is proposed for identification of maize leaf diseases in complex field backgrounds. Furthermore, Deployed on smartphones,	99.59% on PV dataset, the real-time disease recognition on mobile platforms achieves 86.91%,
[53]	The proposed lightweight Lite-MDC architecture uses the MultikernelDepthwise separable Convolutions (MDsConv). The MDsConv module captures spatial features across various scales while maintaining a lightweight design. Reduces computational complexity, employing only 2.2 million parameters, which is a 62% reduction compared to the standard VGG16 architecture. Pre-processing eliminates blurry images and the images are cropped to 1024 × 1024 resolutions and annotated by computer vision experts	Assessed on a total of four datasets, comprising the newly introduced pigeon-pea dataset, along with three publicly accessible datasets. novel pigeon-pea dataset 94.14%, Plant Village dataset 99.78%, Cassava dataset 86.4%, and apple leaf dataset 97.2%

Hyperspectral laboratory Images classification with partial least square-discriminant analysis (PLS-DA) yielded 53% accuracy for downy mildew grapevine disease [23].

A classification accuracy of 90.26% was obtained for Fisher linear discriminant analysis (FLDA) when two features based on continuous wavelet analysis namely optimal

Table 5 Layered CNN

Citation	Proposed Work and Dataset	CNN Model and accuracy
[37]	Multilayer CNN—Comparing results with the other state-of-the-art approaches Particle swarm optimization, SVM, and RBFNN—Mango Leaf—1 Disease—1070 realtime images collected in J&K	MCNN 97.13, PSO 88.39, SVM 92.75 and RBFNN 94.20
[55]	Nine Layer DeepCNN—.54,305 images 13 different plant leaves 39 different classes of PVdataset	NL-Deep CNN 96.46%
[91]	An optimized nine-layer (ONL)-CNN-based plant leaf disease classification (PLDC) model for ensuring accurate results.—using A-FCM and HLCSO(Hybrid leader cat swarm optimization)—Kaggle Dataset -87 K images of 10 plants 38 classes	HLCSO-ONL-CNN 96.45%

features abnormality detection (OF-AD) and optimal feature stress discrimination (OF-SD) from laboratory collected hyperspectral images of Tea leaves for 3 diseases [25].

2.1.2 Variational auto-encoder (VAE)

In [92] segmented healthy and diseased regions of leaf images using Fuzzy C-Means (FCM) clustering. From the segmented clusters various texture-based features are extracted using DWT based GLCM by wavelet decomposition of image. Significant features of gray level, color and wavelet were used for classification. The autoencoder classifier operates as deep neural network in an unguided manner and the likelihood of error value is too low than SVM. But the performances achieved were 80% accuracy for 100 images of apple dataset from Kaggle and 79% accuracy for 100 images collected from sugarcane field.

In [18] the proposed light weight tomato leaf disease identification network using deep DCNN supported by VAE, has increased identification accuracy and achieved accuracies of 98.27% for healthy and 94.17% for diseased leaf identification. They have used VAE network + classification network only for training the dataset and not in the testing phase. Therefore additional calculation and storage was not required in model deployment and production. Gaussian sampling enhances the generalization of the model and reduces overfitting.

2.1.3 Self organizing maps (SOMs)

Multispectral images of detached tomato leaves inoculated with the fungus *Botrytis cinerea* and mock infection solution [19] were captured at regular intervals and fed to an iterative self-organizing (ISO) classifier to separate the masks of the image into two classes with the detection percentage for various image bands ranging from 35 to 63%. The use of feature extraction and feature optimization before classification could improve accuracy.

2.1.4 Modified recurrent neural network (MRNN)

Disease classification with the MRNN, where the standard deviation-based grasshopper horse herd optimization (SD-GHHO) technique was used for segmentation and optimizing the weight of the RNN with severity computation improving the accuracy for 3 standard datasets and one real-time dataset [20].

2.2 Supervised approaches

Leaf images labelled as healthy or with disease names are used for training the algorithm. The input dataset will be separated into training and testing data. In most of the works 70% or 80% will be labelled and the remaining 30% or 20% are tested which will include healthy and all categories of disease samples for effective learning. A research work in [28] separates dataset into 50–50. ML algorithms like NB, KNN, SVM, DT and RF and DL algorithms like RPN, MLP, CNN, and BPNN are supervised approaches found in various research works of Leaf Disease Identification.

2.2.1 Naïve bayes (NB)

In their experimental work [22] used NB with accuracies ranging from 61 to 90% for grapevine leafroll disease (GLD) from field data comprising visible and infrared images from two years and six seasons and planned to focus on early stage detection in the future.

The GLCM and GLRLM feature extraction techniques were applied for various plant images of the Mendeley dataset by [21] and the accuracy achieved by the NB classifier was 84.69%

NB is suitable for binary classification because it requires conditional independence of disease features. The presence of symptoms of more than one leaf disease reduces the accuracy and it is not chosen for the classification of more than one disease.

2.2.2 K nearest neighbour (KNN)

The authors in [16] performed classification with weighted k-nearest neighbour (W-KNN) which yielded accuracies of 93.8%, 87.0% and 84.8% in experiments with three datasets. Tomato leaf diseases were classified via KNN by [61] reported 78.76% accuracy (training accuracy of 80%) when Sobel edge detection, GLCM, Gabor and color feature extraction were used.

From the ImageNet dataset 100 features were extracted from each cherry leaf image via the scale-invariant feature transform (SIFT) algorithm. Classification done via KNN and achieved an accuracy of 96.2% [27]. The work in [28] reported 76.34% accuracy for KNN classifiers which was the lowest in their experimental setup in comparison with SVM and BPNN.

In [93] classified 3 diseases of maize leaf from a dataset using 7 CNN models for feature extraction and KNN classification resulting in accuracies such as AlexNet KNN 93.3, VGG16 KNN 76.7, VGG19 KNN 88.3, ResNet50 KNN 75, ResNet101 KNN 80, GoogleNet KNN 83.3 and Inception-V3 KNN 80.

[23] performed hyperspectral laboratory image classification with KNN which resulted in 66% accuracy for downy mildew grapevine disease [94]. classified cucumber leaf 5 diseases for a dataset from private reference with fine KNN giving 94.6% and subspace KNN giving 94.5% when 150 features of HOG and 100 features of LBP were taken at ten-fold cross-validations.

2.2.3 Decision tree (DT)

Diseases of citrus fruits and leaves using Ensemble Boosted Tree (EBT) resulted in 94.5% accuracy, and DT resulted in 94.5% accuracy which decreased to approximately 85% accuracy when tested with two other datasets [16]. Maize leaf 3 diseases were classified

from a dataset with decision tree and CNN feature extraction resulting in accuracies between 83.3 and 65 [93].

Cucumber leaf 5 diseases for the dataset from private reference with Bagged Tree giving 94.0%, Fine Tree giving 82.0%, Boosted Tree giving 66.20% and Medium Tree giving 59.40% when 100 features of HOG and 100 features of LBP taken at tenfold cross-validations were explored [94].

2.2.4 Random forest (RF)

The authors experimentally classified citrus diseases from a dataset collected manually with the help of experts and the citrus research centre by using RF with a classification accuracy of 76.8% [67]. An accuracy of 96.49% was achieved by the RF classifier when the GLCM and GLRLM feature extraction techniques were applied to various plant images from the [21]. They have not tried on-field images with their approach.

In an experiment with pre-processing, HOG, GLCM feature extraction PSO feature selection and RF classification, 96.1% accuracy for 5 plants and 3 diseases from images of the PV dataset was obtained [95]. The classification of hyperspectral tea leaf images with 90.69% accuracy for 3 diseases was reported in [25] but their method has not been applied to field or laboratory digital images or has not been extended to other diseases or other plants.

In [24] features were extracted via CNN. RF was applied for the classification of banana leaf diseases and the accuracies of different methods were VGG 16 RF 98.85%, ResNet RF 97.14%, MobileNet RF 98.85%, InceptionResNetV2 RF 99.42%, Inception RF 98.28%, Xception RF 100%. They have planned to enhance the work with the inclusion of abiotic stress with real-time cloud processing.

2.2.5 Region proposal network (RPN)

In a research work [26] identified three diseases in leaves by training via RPN and the Chan-Vese algorithm and classified the leaves via transfer learning the model has overcome the environmental complexity and improved accuracy to 83.57%. However as the Chan-Vese algorithm requires repetitive calculations and runs for a longer time, the authors planned to use a Neural Network in its place to speedup training and end the iteration faster.

2.2.6 Back propogation neural network (BPNN)

Transfer learning techniques are applied to fine-tune the CNN from the pre-trained GoogLeNet on 100 features of the ImageNet dataset [27]. One hundred features are extracted from each cherry leaf image via the scale-invariant feature transform (SIFT) algorithm and classified with 97% accuracy via BPNN.

The impact of stress type on five different paddy crop varieties during the booting growth stage was evaluated with field images via BPNN classification which yielded an accuracy of 89.12% which was higher than that of the SVM and KNN methods in [28]. The feature selection was performed via the sequential forward floating selection (SFFS) algorithm. The effects of environmental conditions such as extreme temperatures and soil factors, the presence of combinatorial stresses, the quantification of stresses, and the prediction of the gap between yield potential and yield under stress are the factors proposed by the authors for further studies.

2.2.7 Support vector machine (SVM)

A novel method in [16] proposed segmentation (weighted segmentation and saliency map combined with Image fusion/morphological operations) and multiclass SVM (M-SVM) classification for citrus fruit and leaf 6 diseases while comparing the proposed segmentation method results with those of the EM, Otsu, and active contour methods on complex images. Three types of features extracted are combined via a simple concatenation based fusion method to form a codebook, which was fed to PCA for feature selection. They experiment with the Citrus Image Gallery dataset which achieves 95.8% accuracy, a combined dataset with 89.0% accuracy and their own local dataset with 90.4% accuracy. The construction of a DL model that needs a large citrus dataset is planned for future studies.

Automatic disease detection for 3 diseases on images of vine leaves taken on smart phones or tabs was trained via one class SVM [96]. Testing was performed on a variety of crops with an accuracy of 95%. The grabcut algorithm for segmentation and LBP for feature extraction were used.

Three apple leaf diseases were classified with one-vs.-all-M-SVM classifier and a genetic algorithm was used to optimize the extracted features [97]. The classification was also attempted for tea leaves which yielded 95.04% accuracy when less than 2.3 M parameters were used in 10 ms.

In [93] classified maize leaf 3 diseases from PV dataset via 7 CNN models for feature extraction and an SVM for classification. The accuracies obtained were 95% for AlexNet SVM, 90% for the ResNet101 SVM, 88.3% for VGG16 and VGG19 SVMs, 86.7% for the ResNet50 and Inception-V3 SVMs, 83.3% for the GoogleNet SVM. The authors have planned to increase the percentage of accuracy by adding optimization methods in CNN architectures.

An Ensemble nonlinear SVM was used for multiple crop leaf disease classification [98] which was more accurate than the KNN but less accurate than the CNN. However the computational overhead of the CNN is high: hence the ENSVM has optimal performance.

Images of 12 plants collected from Katra University were classified with two class SVM and multiclass SVM by [29]. They have done segmentation based on thresholding and feature extraction was performed with LBP and HOG and achieved 99% accuracy. However they haven't tried the methods for large datasets.

Classification with a Bayesian optimized SVM [95] gave 92.2% accuracy for 5 plants and 3 diseases from images of the PV dataset. In [27] classified via SVM and achieved 95.5% accuracy. The authors of [28] in their experimental setup for field images achieved average stress classification accuracy of 84.44% with the SVM.

2.2.7.1 Kernel functions In [99] the fusion of superpixel clustering, K-means clustering and pyramid of histograms of orientation gradients (PHOG) algorithms were proposed for classification via a context aware support vector machine C-SVM with a radial basis kernel function for 3 diseases of apple and cucumber which resulted in accuracies between 82 and 92%.

The SVM cubic kernel with 98.4% accuracy for 4 diseases of different plant images segmented with K means clustering and extracted features via the GLCM and LBP [100]. They have planned to extend the database for more plant disease identification and use a

large amount of data for training to improve accuracy. An SVM with a quadratic kernel function for 3 crop 3 diseases in dataset images and fuzzy histogram equalization during pre-processing yielded 88.1% accuracy [101]. The authors concluded that the accuracy of disease detection can increase as the number of training samples increases and that the SVM settings with other kernel functions are not tested.

Cucumber leaf 5 diseases for a dataset from private reference with SVM cubic resulting in 91.90%, 91.60% and 91.80% accuracies in under three experimental conditions by varying the number of features [94]. The HOG, LBP, and color feature vectors are fused via feature scoring to obtain the optimal features. They have not applied feature optimization techniques and have not tested their classification methods on field images and hyperspectral images.

Three diseases of rice plants were classified via radial basis function SVM (RBF-SVM), SVM, random forest, and ID3, [102]. Images are pre-processed with the use of a mean filter and histogram. The k-means technique segments the data into several groups. PCA extracts features from the data. The accuracy of the RBF-SVM is high according to the experimental results.

SVMs with four kernel functions namely, linear, polynomial, radial basis function, and sigmoid were experimented with four feature extraction (1) GLCM, (2) HOG, (3) LBP, and (4) Harris, BRISK, ORB, MSER, and Hough (line length and point number) for classifying 3 diseases of grape leaves in the PV dataset [30]. SVM linear function with the GLCM feature yielded the highest accuracy of 98.71% which is greater than that of the DL methods in terms of reduced time complexity because of segmentation by K-means clustering of diseased areas by selecting ROIs automatically by thresholding. PCA and relief selection identify the best and most important features for diagnosing the disease.

The hyperspectral imaging in [31] detects crown rot disease for 4 varieties of wheat plant via one-class classifier of SVM with the radial basis function (RBF) kernel. The experimental results show successful diagnosis of infected plants in a greenhouse approximately 30 days after infection when there were no visible symptoms on shoots. After the plant segmentation the data is smoothed with Savitzky-Golay filter to remove random noise created by damaged cells of detectors. In five-fold cross validation the accuracy is more than 99.98%.

2.2.7.2 Clustering techniques for segmentation A rule-based semiautomatic system using the concepts of k-means is designed and implemented for binary classification of soybean leaf images with complex backgrounds collected from the PV dataset. Furthermore, 3 diseases are classified via SVM in [32]. K-means clustering segmentation in the proposed system uses 7 rules framed after closely observing the formed clusters and dividing the leaves into 3 clusters. As the diseases considered in this work vary in color thresholding approach was not used in k-means segmentation. Moreover a severity estimation module uses the clusters for computing severity. Nine feature extraction methods were tested and the top three average accuracies obtained were 84.3% for color + texture features, 83.59% for color + Gabor features and 82.52% for color + Haralick features. Three models with varying training percentages of 50, 60 and 70 for SVM classification were tested which confirmed the usual observation of increase in accuracy with increasing in training percentage. The proposed system has to be tested for other plants with extended diseases and real time images.

The SVM is trained with different algorithms for tea leaf disease detection and the accuracy is compared in [103]. The method that used simple linear iterative cluster (SLIC) with SVM was found to be effective and yielded higher accuracy. Five tea leaf diseases were classified with multiclass SVM in [104]. First image segmentation with a multiobjective clustering technique based on NSGA-II was used and features were extracted via PCA.

Reference [25] is one of the article describing experiments where classification was performed with SVM for hyperspectral tea leaf images with 90.67% accuracy. They have initially performed k-means clustering and abnormal area detection with SVM and then identified 3 diseases using a second SVM classifier: however their method has not been applied to field or laboratory digital images or has not been extended to other diseases or other plants.

2.2.8 Feed forward neural network (FFNN)

Brinjal plant four diseases were identified from field images via a camera with the FFNN and cascade feed forward neural network (CFNN) along with pre-processing, segmentation and feature extraction and obtained accuracies as of 91% for the FFNN and 98.1% for the CFNN [33]

2.2.9 Multilayer perceptron (MLP)

Papaya leaf 3 diseases were classified via MLP while using k medoid clustering for segmentation, and the backpropagation algorithm for training [105]. Accuracy was calculated for segmentation and classification and has planned to detect all diseases of papayas in future.

[35] has classified only Hyperspectral images of Tomato leaves were classified into 4 classes healthy, asymptomatic, early and late disease development stages via the MLP with accuracies ranging from 99 to 100 by testing with different vegetative indices and hyperspectral wavebands without pre-processing and feature extraction while identifying 5 vegetative indices from 35 vegetative indices as best.

An apple dataset with 106 images was classified for two diseases via MLP and GLCM feature extraction [34]. The gradient descent back-propagation algorithm is used to optimize the weights. Accuracy 99.1% was achieved for the above dataset, but it was not tested for a large dataset.

The GLCM and GLRLM feature extraction techniques were applied to various plant images of the Mendeley dataset by and the accuracy achieved by the MLP was 98.32% [21]. Hyperspectral laboratory image classification with MLP achieved 81% accuracy for downy mildew grapevine disease [23], whereas an increase in classification accuracy could have been achieved with feature extraction methods.

2.2.10 Radial basis functional neural network (RBFNN)

The classification in [37] resulted in 94.20% accuracy with the RBFNN for a single disease of mango leaf for a real-time dataset consisting of 1070 images. The detection of 2 diseases of rice plants with a low-complexity RBFNN classifier in [36] yielded 95% accuracy whereas PCA and the Gabor filter were applied to the images before classification.

2.2.11 Convolutional neural network

A CNN is a deep learning network design that learns from the input without the requirement for human feature extraction [98] viewed a CNN as a type of MLP with computational overhead that performs convolution at the beginning and deconvolution at the end to integrate the results which benefits the hierarchical pattern of the leaf data.

Using existing CNN models in ML/DL libraries might be helpful in some cases. However to achieve better control and understanding, CNN must be built and implemented. A CNN typically consists of several layers but the following layers are commonly observed:

2.2.11.1 Convolutional layer

- Filters (kernels) are applied to the input image to extract features such as edges, textures, and colors.
- The output is a feature map that represents the detected features.

2.2.11.2 Pooling layer

- The dimensionality of the feature map is reduced while important information is preserved.
- Common pooling techniques include max pooling and average pooling.

2.2.11.3 Fully connected layers

- Like traditional neural networks, these layers process flattened features and perform the final classification.
- The output layer typically uses a Softmax activation function to produce probabilities for each class.

In addition three additional layers can be found as follows.

2.2.11.4 Flattening layer The 2D feature map is converted into a one dimensional array to be fed into the fully connected layers.

Cucumber leaf 4 diseases were classified via a deep CNN for images segmented and augmented from the PV dataset as well as captured images in Ma et al. [106]. The Accuracy achieved by the DCNN on the unbalanced dataset was 93.4% and that achieved by the DCNN on the balanced dataset was 92.2%

Apple leaf 2 diseases from hundreds of images available from a research institute was classified by [107] using ROI-aware DCNN with 84.3% accuracy. The ROI-aware DCNN was built along with two subnetworks, one for identifying ROI and one for disease classification.

The authors in [108] proposes CNN from baseline ResNet 50 architecture that integrates image and categorical data and also generated a challenging dataset of more than one hundred-thousand images taken by cell phone in real field wild conditions that contains almost equally distributed disease stages of seventeen diseases and five crops (wheat, barley, corn, rice and rape-seed). Images with several diseases and also with similar visual symptoms were also included in their experiment and achieved balanced accuracies in the range of 55% to 99%. Diseases with a lower number of images

are benefited of the use of a multi-crop model. Multiplicative suppression on the image embedding layer causes gradient inconsistencies and vanishing leading to poor convergence and needs additional focus and analysis.

In a work in [109] the authors classified tomato plant 1 disease with Mask RCNN and Masked Region based CNN with Augmentation, Segmentation of infected region using CNN (U-net and mask RCNN) and achieved accuracies of MRCNN-ResNet50 81.01, MRCNN-ResNet50 (A) 85.67, MRCNN-ResNet101 81.09, MRCNN-ResNet101 (A) 82.72.

In [38] presented automatic plant leaf disease detection via a deep convolutional neural network (DCNN) to increase the feature representation and correlation and generative adversarial network (GAN) for data augmentation to address the data imbalance problem with 99.74% for 9 diseases in tomato plant taken from the PV dataset: however this method does not addresses less generalization.

In [39] the authors proposed regional-based convolutional neural networks and U-Net (CRUN) for segmenting 2 leaf diseases of banana and sugarcane from the augmented leaf dataset. The convolutional and max-pooling layers are highly utilized in the CRUN's U-Net architecture. The segmented images are subsequently subjected to a morphological process to identify the level of disease in the leaf for banana, sugarcane and multi-class. CRUN performances are evaluated using the traditional classification performance metrics of accuracy, sensitivity and specificity. The real time images gave 98 and above accuracy whereas the Kaggle dataset of sugarcane, banana and multiclass reached 99 and above accuracy.

2.2.12 Pre-trained CNN models

Pre-trained CNN models are pre-trained on massive image datasets such as ImageNet. They have learned to extract meaningful features from images, resulting in high accuracy with minimal training data. They can be used from popular deep learning frameworks such as TensorFlow/Keras and PyTorch. These models can be utilized because they can be fine-tuned for leaf disease datasets. Layers can also be modified, or new layers can be added and trained from scratch as the leaf disease dataset is entirely different from that of the ImageNet dataset used for training. This is known as transfer learning. If we perform fine tuning instead of transfer learning for pre-trained models, the accuracy will be affected. Performance also drops when real-world datasets are used in Table 2 [42] and Table 3 [52, 81]. The reasons are due to one or many factors discussed in 1.3 Review highlights.

Some research works have modified the CNN layers or designed new CNN and have provided the classification results. Some have been given separate names also. They are presented in Table 4 and the works of multi-layered CNN are presented in Table 5.

2.2.13 Attention network

Both-channel residual attention Network model (B-ARNet) was developed based on the basis of ResNet in [57]. ABCK-BWTR and B-ARNet with 89% accuracy on tomato Plant 5 diseases trained on field images and tested on the PV dataset. The authors have suggested including more images of tomato leaf in the dataset to improve the recognition effect under complicated background.

A fine-grained disease categorization method based on an attention network to classify PV dataset images of 2657 peach leaves in 2 categories and 18,162 tomato leaves in 10 categories resulting in 92–95% accuracy was proposed in [110]. The model can be integrated into smart phone and unmanned aerial vehicles and have to be extended to other crops or application scenario.

A group multiscale attention network (GMA-Net) was proposed in [56] with group multiscale dilated convolution (GMDC) module for multiscale feature extraction and a cross-scale attention feature fusion (CAFF) module for multiscale attention feature fusion for rubber leaf 5 classes of 2,788 laboratory images for training (98.06%) were used and the results were verified with 54, 303 images from the PV dataset with 99.43% accuracy. The model reduces the influence of complex background in the image and the authors have planned to deploy the model in the mobile devices.

To achieve better performance in the fine-grained classification of the 59 crop diseases, a modified lightweight convolution neural network SqueezeNext(with multiscale convolution and coordinate attention) with 91.94% accuracy was proposed in [111] for the AI Challenger 2018 crop disease detection dataset after resampling to address the data imbalance issue. They have used three different methods early stopping, data augmentation and weight decay for preventing over fitting. The images are normalized to the size 224×224 and data augmentation was done by applying random transformation. The model in 2.83 MB is suitable for deployment on mobile terminals and other embedded resource-constrained devices, which is only 3.45% of the Xception (82.03 MB),but has not been fully evaluated, especially in a real field environment.

2.2.14 Capsule neural networks

An optimized capsule neural network (CapsNet) to detect and classify 10 tomato leaf diseases via standard dataset images by mitigating overfitting with data augmentation and pre-processing techniques during the training phase was utilized in [112]. CapsNet was chosen over CNNs because of its superior ability to capture spatial positioning within the image. Accuracy was not revealed and so the classification methods' success cannot be measured. For major plant diseases, symptoms can be found in any area of the leaf. Only some diseases occur at the tip, edge, bottom, or the middle portion of the leaf. A capsule structure in CapsNet preserves the position and orientation of features in the image. In a face recognition domain, if the features of the eyes, nose and mouth are placed in the wrong position, CapsNet can identify and mark it as 'Not a Face'. In LDI, this structure would add extra computations that are not required.

2.3 Self-supervised approaches

Classifications with supervised approaches have good accuracy but are expensive because of training with labeled data. Unsupervised approaches do not consider the use of manual annotation, so it is difficult for the model to guarantee attention to informative regions, which usually results in relatively low classification accuracy.

2.3.1 LFC-Net

A novel model, which consists of 3 networks, including a location network, a feedback network, and a classification network, named LFC-Net with self-supervision mechanism was proposed in [58]. This model was tested on three datasets namely, random selection

373, PV and data augmentation and has achieved accuracy of 99.7% for 8 diseases in tomato plants.

2.3.2 Vision transformer (ViT)

A vision transformer-based lightweight model, ConvViT uses convolutional structures, transformer structures and attention mechanisms to extract effective features of crop disease spots and classify apple leaf diseases for self-acquired images with 96.85% accuracy [60].

A standard model based on transformer and self-attention was proposed and achieved 98.7% accuracy for Maize plant 4 diseases when tested with 7,701 RGB images from experimental fields and a public dataset [113].

In experiments with 14 CNN and 17 vision transformer models in [59], deep learning demonstrated high accuracy in distinguishing grape leaf 3 diseases and recognizing 5 varieties of grape leaves. Notably, four models (Swinv2-Base-Win8, Deit3-Base, Maxvit-Small and Maxvit-Large) achieved 100% accuracy on PV dataset Swinv2-Base achieves 100% in classification of leaf images in PV dataset and recognition of leaves in Grapevine dataset.

2.4 Other approaches

2.4.1 Statistical

Tomato leaves were classified as healthy or unhealthy via KNN and unhealthy leaves were classified via a probabilistic neural network (PNN) with Accuracy 91.88% [61]. Sobel Edge Detection, GLCM, Gabor and color feature extraction were used.

2.4.2 Fuzzy

In [62] an adaptive neuro-fuzzy inference system (ANFIS) architecture containing 5 layers with gray-level spatial dependence-based feature extraction for plant disease recognition by classification (GLSMPDR) and achieved accuracies of 90.7% and 98.0% for the TPDS 1.0 and BPDS 1.0 datasets respectively created for 4 diseases of tomato and brinjal.

The robust feature extraction approach combines the GLCM, moment complex Gabor and curvelet transform-based features with neuro-fuzzy classifier and achieves 93.18% accuracy for the PV dataset [114].

Galls present on the leaf of *Alstonia Scholaris* (Medical plant) were identified with the Fuzzy Based Function Network enabled with IOT by [110]. Acquired images are pre-processed and features are extracted using the Scale-invariant feature transform (SIFT) method and optimized with FireFly algorithm. Specificity, Sensitivity was measured but classification accuracy was not calculated.

2.4.3 Hybrid models

CNN which was preferred for automatic feature extraction and RNN which was preferred for holding information for long periods using LSTM was combined in [115] to identify tomato leaf 9 diseases of PV dataset and obtained 98.25% as the best categorical accuracy and accuracy of 81.75% for disease classification. The parameter EPOCHS have been varied and tested for two different time steps but other parameters were not used.

A hybrid model of CAE combined with CNN is proposed in [63] for 4457 leaf images of peach plant bacterial spot disease from PV dataset. CAE reduces dimensionality by providing compressed domain representation of input images to the CNN from 256×256 to 32×32 while reconstruction loss is minimized with the usage of normalized root mean squared error (NRMSE) loss function. The model has achieved 98.38% accuracy and has used thousands of training parameters when compared to millions of training parameters by an individual model.

In order to increase robustness with different disease severity [62] have proposed a fusion network based on CNN ResNet18 and two-layer graph convolutional neural network (GCN) for tomato and cucumber six diseases consisting of 1,715 images and a static graph structure dataset describing the disease symptoms in text. They have obtained 97.62% accuracy with the fusion model and suggest using the dynamic graph neural network training method for optimization.

Multiclass-SVM was used for classification of tomato plant diseases into 6 classes on training with 600 images from crowdAI plant village challenge. They have also done implementation for disease classification and stages of disease for the dataset from GKVK, Department of agriculture, Karnataka [116]. Image segmentation with K-means clustering and feature extraction with the GLCM was performed. The classification results are compared with those of Alexnet, SVM Alexnet-SVM, Alexnet-NaiveBayes. The accuracies were—92.4%, 75%, 87.3% and 80% respectively. Remedies were obtained on the basis of the classification results given by the Alexnet model. The authors have suggested using the model for one of the most sensitive and costly treatments in India in terms of leaf concern are that of pomegranate as the expertise in this field is rarely available.

3 Limitations and discussion

The review briefly describes existing research works and there is no thematic or quantitative synthesis but reiterates limitations of the field. From the review we can observe that the laboratory images or images taken on controlled environments with plain background were the most widely used type of dataset. Some researchers have collected images in real environment using digital cameras or other capturing devices. The segmentation process has to be included to separate leaf portions from the background. In many works involving real images the collected real images are mostly in hundreds and represent few diseases. Most of the research was not carried out on multispectral images. Accuracy of some classification methods has dropped when real images were taken for input.

LDI for and plant needs standard and complete labelled datasets which should possess images of.

- Healthy leaves from tender shoot to fully grown stage and to trash dry leaves.
- Diseased leaves from early stage to fully developed stage.
- Multiple diseases in one leaf.
- All possible background.
- All possible distortion (dust, mud, worms etc.)
- Acquired from various distances, angle and light effect.

Such a dataset doesn't exist [117]. Available standard datasets like PV, Kaggle and Cassava have large collections of images for some plants and some diseases with few farm images. Some researchers have collected images on their own for some plants and diseases. Some of the images of plant leaves and their diseases are not found in these datasets.

The descriptions of the visible symptoms of various plant diseases and their relationship with the pathogenic groups are given in [117]. Apart from accuracy other agronomic context like field conditions, disease symptoms and practical deployment constraints in the existing works were not included in the review. Also superficial agronomic problem framing was not focussed in the review.

This work focuses on the advantages and disadvantages of various feature extraction techniques and the importance of feature engineering to identify and select relevant variables from the raw data. The results of the experiments in [30] attest to the role of various feature extractions in influencing the classification accuracy. They also compared the classification accuracies with and without the segmentation backing the role of segmentation. The support for segmentation can be found in the segmentation accuracy results in [16]. The three experimental setups in [97] calculated the classification accuracies without feature selection, PCA feature selection, and genetic algorithm-based feature selection. The increased classification accuracy is the affirmation of the role of feature selection.

On review of existing classification approaches for LDI number of research works available under unsupervised approaches was few in number. Real time images were not tested with unsupervised approaches. Variational autoencoder (VAE) has not been tried for visible images in real environment or other spectral images like hyperspectral or IR images. ISO classifier was not performed on visible spectrum of laboratory or real images.

Existing and evolving supervised approaches were tried with three types of dataset and results were recorded. All the supervised algorithms require high end computing time and memory as the training data should include many leaf samples covering all stages of chosen diseases in all capturing environments.

CNN was the most used classification approach and has also shown invincible high accuracy in copious research works. Despite their computational complexity, it was chosen for its high accuracy, automatic feature extraction by the convolutional layer, and its ability toward multi-classification for many classes. Through augmentation techniques and transfer learning, LDI with CNN can be extended to more plants for multiple diseases. From Table 1, the approach has shown robustness low accuracy drop towards real time images, but has never been tried for images from invisible electromagnetic spectrum like hyper-spectral, IR, or microscopic images. In the same experimental setup, the accuracy of the CNN is higher than other ML methods [54]. The rationale for selecting CNNs for leaf disease is mainly, because each layer of the convolution layer in the dense block is small, only 1×1 and 3×3 , each convolution kernel is responsible for learning the tiny features [86]. Due to the ambiguity, complexity, and similarity of leaf diseases, as well as the complex background interference often present in disease images collected in real-time scenarios, general disease recognition models face problems such as poor generalization and low accuracy in practical applications. [90] CNN provides various

adjustable parameters like the dropout rate, the learning rate used in complex processing for solving classification and pattern recognition problems [37].

Other DL supervised approaches like RPN, BPNN, FFNN, MLP, Attention Network, and Capsule Neural Networks were explored by few researchers on a limited number of images of either single or selected plant or diseases. The performance was not fully evaluated for images affected by the lighting discrepancy, exposure level, and background. Deep Learning approaches use traditional data augmentation methods such as rotation and flipping which are limited in their effectiveness when faced with imbalanced datasets. Advanced data augmentation techniques, including C-DCGAN [42], DoubleGAN [49], CycleGAN [52], and LeafGAN [70], increase the training data and provide samples for classes where diseased leaf images were not available. A novel approach for augmentation, namely, object-based neural style transfer [118] generates synthetic images that more accurately represent underrepresented classes and enhance the CNNs ability to generalize across different classes. A separate review based on the work of CNN pre-trained models alone for LDI as in [119] helps analysing CNN models in depth. As new DL models were emerging their performance can be evaluated for same dataset as in [120] and the classification have to be extended to many diseases for multiple plants for increased generalisation.

SVM was the only classification method applied to all the three type of images with good accuracies. The SVM takes the data to a higher dimension and classifies it without actually transforming the data into a higher dimension by using a kernel function. This classification is suitable for nonlinear and overlapping data like features obtained from different classes of leaf images. SVM performs this classification by utilizing a small training dataset and is therefore suitable for all three types of datasets in LDI. A kernel function in the SVM identifies the mathematical function to take the data to a higher dimension for classification. It helps to identify the hyperplane or hypershape that classifies the data with reduced features.

The traditional ML classifiers like KNN, NB, and Decision Trees are experimented with the same dataset in an explorative study performed in [16] and [97]. The results have showed that the classification accuracy of the SVM is superior to these classifiers and are highly dependent on the methods of feature extraction and optimization steps before classification [24, 28] Fig. 6 shows the supremacy of SVM over these traditional classifiers in terms of NOP where these classifiers were used. Figures 3, 4 and 5 further sustain the accuracy of the SVM with only one exception of [24] which shows RF, which is an ensemble of decision trees performing better. The lowest and highest accuracies from Table 1 under the six groups namely Unsupervised, Supervised SVM, Supervised CNN, Supervised Others, Self-supervised and Other approaches are shown in Fig. 3 for laboratory dataset, in Fig. 4 for real dataset and Fig. 5 for hyperspectral dataset. The group, supervised others include all the supervised classification methods other than SVM and CNN. They are NB, KNN, DT, RF, RPN, BPNN, FFNN, MLP, Attention Network and CapsNet (Fig. 6).

The performance of the individual classification methods has been reviewed from the works of existing literature in this study. The framework to in Fig. 7 categorize and compare ML/DL/Hybrid methods in LDI would help readers better understand the landscape and identify gaps. Table 6 shows the performance of general approaches in classification in terms of accuracy, training time, dataset size, robustness, and computational

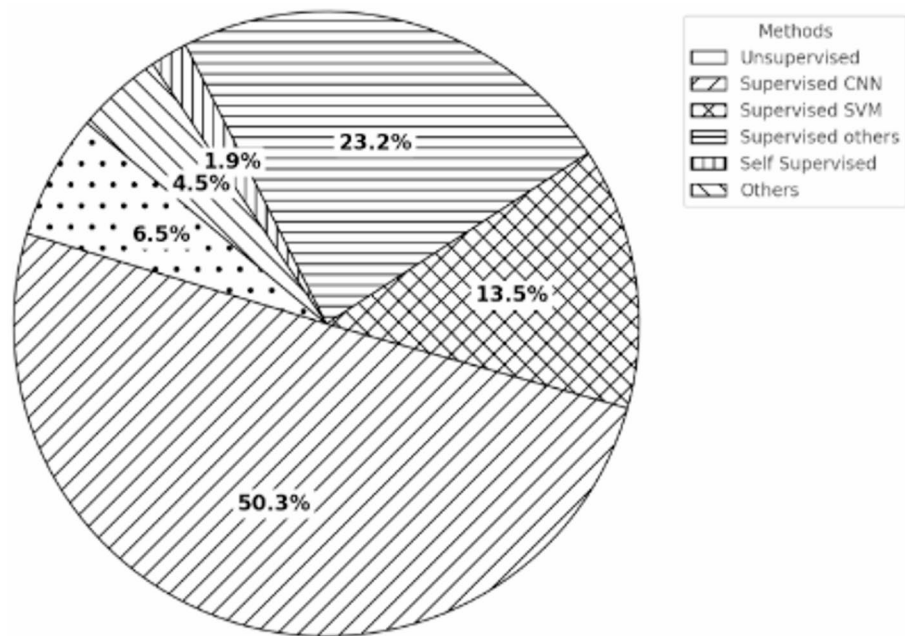


Fig. 6 Distribution of classification methods across papers reviewed

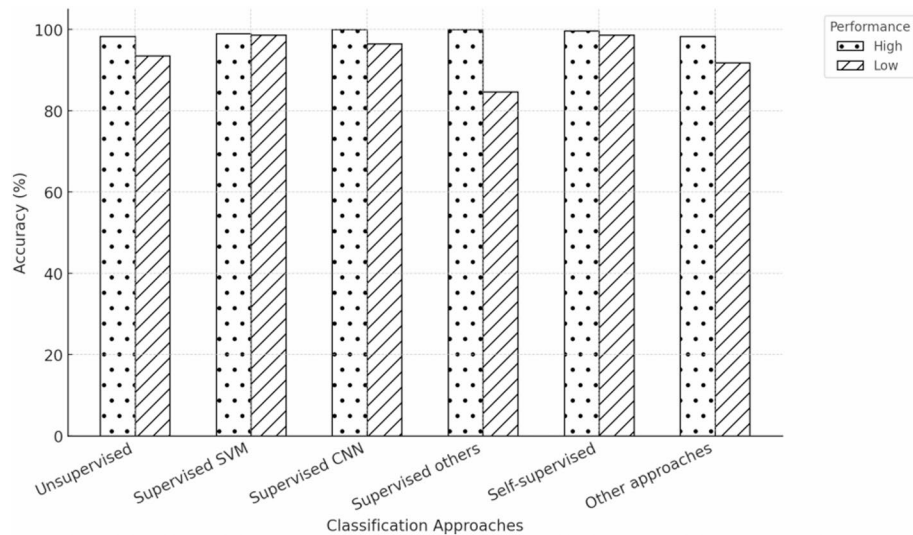


Fig. 3 Highest and lowest accuracies from Table 1 for various approaches on laboratory dataset

requirements. The overall performance of the individual classification methods has been presented in that table under the three broad categories of unsupervised, supervised and self-supervised. The novel methods under Sect. 2.4 other approaches have to be analyzed separately and not under the broad categories in Table 6. In the hybrid approaches presented under 2.4.3 hybrid models, the researchers have tried to combine the benefits of individual models and tried to improve accuracy and minimize loss while maintaining optimum training parameters. The hybrid model in [115], which combines the boons of CNN and RNN can also be helpful in performing LDI along with severity estimation simultaneously, will be a consideration for future work in smart agricultural technologies. A systematic review with well-defined exclusion criteria, quality assessment of

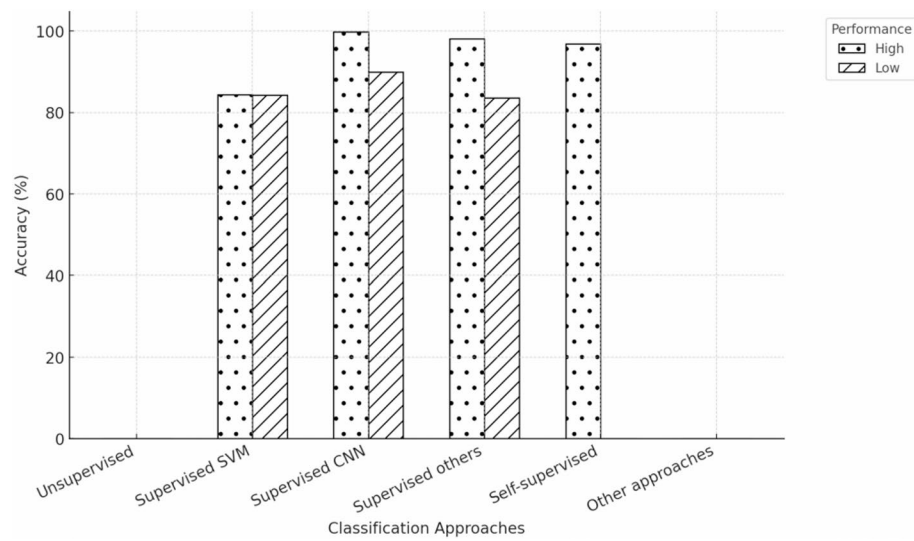


Fig. 4 Highest and lowest accuracies from Table 1 for various approaches on real images

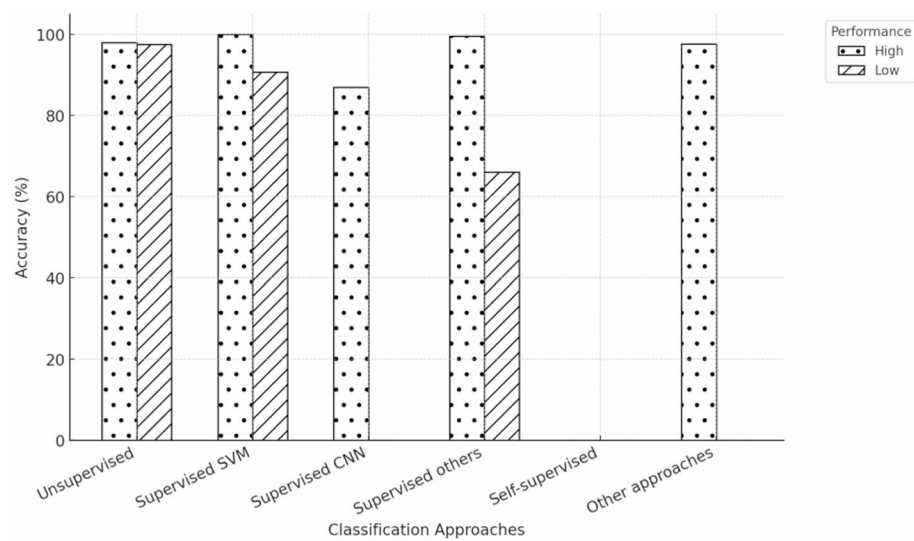


Fig. 5 Highest and lowest accuracies from Table 1 for various approaches on hyperspectral images

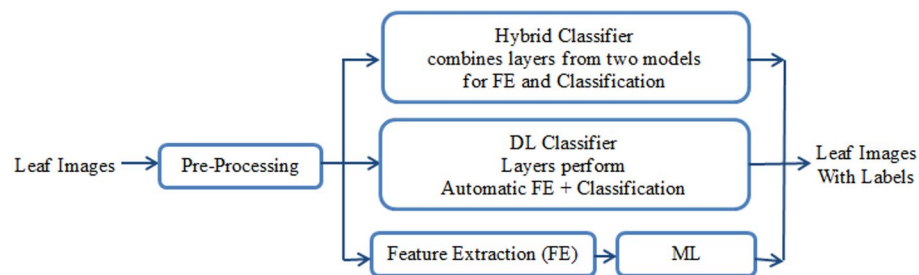


Fig. 7 Frame work for ML/ DL/ Hybrid methods in LDI

Table 6 Performance across supervised, unsupervised, and self-supervised approaches

Approach	Accuracy	Training Time	Dataset Size	Robustness	Computational Requirements
Unsupervised	Low	Medium	Small	Good	CPU, GPU, (TPU*)
Supervised	High	High	Large [#]	Poor	
Self-supervised	Medium	Low	Medium Large	Average	

[#]SVM is the exceptional supervised method that works well even for small training dataset

*TPU, Tensor Processing Unit designed for high volume of low precision computation for neural network machine learning

included studies, reproducibility of the review process, critical analysis of classification methods, synthesis of key trends and limitations, would provide future research directions across approaches.

4 Conclusion

This work focused on reviewing the recent approaches in leaf disease identification (LDI) with image Classification methods. A rundown of different types of classification approaches like Discriminant Analysis, VAE, NB, KNN, DT, RF, RPN, BPNN, SVM, FFNN, MLP, CNN, attention network, transformers, fuzzy and hybrid models are presented. The review also includes type of dataset and classification accuracy the main evaluation metrics recently used by researchers in the area. Also the methods of pre-processing, segmentation, augmentation, feature extraction, feature reduction and the usage of optimization algorithms are observed and mentioned as they have impact on the results. Still there is a lot of work to be done in LDI on real time images that will help in agriculture and humanity as a whole. Particularly reviewing with focus on Agronomic concepts would be more useful. This comprehensive review will give a brief introduction to various current research works for classifying leaf diseases and highlighted potential studies for developing LDI systems that can be extended to multiple plant and leaf diseases with increased generalization. LDI challenges are linked to noise removal and presence of multiple disease symptoms simultaneously. The dataset chosen, its type, its amount and the methods involved in the existing research works are mentioned along with the performance and limitations. This will help to identify unexplored ways in LDI.

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Author contribution

1. Andal. Pa —Wrote manuscript, edited manuscript, prepared figures and tables 2. Thangaraj. M—Conceptualized, Supervised and Reviewed.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Informed consent

Not Applicable.

Research involving human and/or animals

Not Applicable.

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References

1. Nagaraju M, Chawla P. Systematic review of deep learning techniques in plant disease detection. *Int J Syst Assur Eng Manag.* 2020;11:547–60.
2. Bhagat M, Kumar D. A comprehensive survey on leaf disease identification & classification. *Multimedia Tools Appl.* 2022;81(23):33897–925.
3. Tripathi A, Chourasia U, Dixit P, Chang V. A survey: plant disease detection using deep learning. *Int J Distrib Syst Technol (IJ DST).* 2021;12(3):1–26.
4. Demilie WB. Plant disease detection and classification techniques: a comparative study of the performances. *J Big Data.* 2024;11(1): 5.
5. Barbedo JG. Factors influencing the use of deep learning for plant disease recognition. *Biosyst Eng.* 2018;172:84–91.
6. Iqbal Z, Khan MA, Sharif M, Shah JH, ur Rehman MH, Javed K. An automated detection and classification of citrus plant diseases using image processing techniques: a review. *Comput Electron Agric.* 2018;153:12–32.
7. Liu J, Wang X. Plant diseases and pests detection based on deep learning: a review. *Plant Methods.* 2021;17:22.
8. Ngugi LC, Abelwahab M, Abo-Zahhad M. Recent advances in image processing techniques for automated leaf pest and disease recognition—a review. *Inf Process Agric.* 2021;8(1):27–51.
9. Gurle AS, Barathe SN, Gangule RS, Jagtap SD, Patankar T. Survey paper on tomato crop disease detection and pest management. *Int J Appl Evol Comput.* 2019;10(3):10–8.
10. Nigam S, Jain R. Plant disease identification using deep learning: a review. *Indian J Agric Sci.* 2020;90(2):249–57.
11. Joshi BM, Bhavsar H. Plant leaf disease detection and control: a survey. *J Inf Optim Sci.* 2020;41(2):475–87.
12. Delfani P, Thuraga V, Banerjee B, Chawade A. Integrative approaches in modern agriculture: IoT, ML and AI for disease forecasting amidst climate change. *Precis Agric.* 2024;25(5):2589–613.
13. Nithiya S, Annapurani K. Biotic leaves disease classification and detection using different classifier analysis in agriculture. 2020.
14. Rampurkar V, Shah K, Todmal M, More A, Lakhotiya I. An approach towards disease detection in potato leaf—a review. *Int J Prog Res Sci Eng.* 2022;3(12):62–6.
15. Vishnoi VK, Kumar K, Kumar B. Plant disease detection using computational intelligence and image processing. *J Plant Dis Prot.* 2021;128:19–53.
16. Sharif M, Khan MA, Iqbal Z, Azam MF, Lali MIU, Javed MY. Detection and classification of citrus diseases in agriculture based on optimized weighted segmentation and feature selection. *Comput Electron Agric.* 2018;150:220–34.
17. Abdulridha J, Ampatzidis Y, Qureshi J, Roberts P. Laboratory and UAV-based identification and classification of tomato yellow leaf curl, bacterial spot, and target spot diseases in tomato utilizing hyperspectral imaging and machine learning. *Remote Sens.* 2020;12(17): 2732.
18. Wu Y, Xu L, Goodman ED. Tomato leaf disease identification and detection based on deep convolutional neural network. *Intell Autom Soft Comput.* 2021. <https://doi.org/10.32604/iasc.2021.016415>.
19. Fahrenttrapp J, Ria F, Geilhausen M, Panassiti B. Detection of gray mold leaf infections prior to visual symptom appearance using a five-band multispectral sensor. *Front Plant Sci.* 2019;10: 628.
20. Sreedevi A, Manike C. A smart solution for tomato leaf disease classification by modified recurrent neural network with severity computation. *Cybernet Syst.* 2022. <https://doi.org/10.1080/01969722.2022.2122004>.
21. Patnaik V, Mohanty M, Subudhi AK. Identification of healthy biological leafs using hybrid-feature classifier. *Imaging Sci J.* 2021;69(5–8):239–53.
22. Sinha R, Khot LR, Rathnayake AP, Gao Z, Naidu RA. Visible-near infrared spectroradiometry-based detection of grapevine leafroll-associated virus 3 in a red-fruited wine grape cultivar. *Comput Electron Agric.* 2019;162:165–73.
23. Hernández I, Gutiérrez S, Ceballos S, Iñiguez R, Barrio I, Tardaguila J. Artificial intelligence and novel sensing technologies for assessing downy mildew in grapevine. *Horticulturae.* 2021;7(5):103.
24. Abisha A, Bharathi N. An hybrid feature extraction and classification using Xception-RF for multiclass disease classification in plant leaves. *Artif Intell.* 2023;37(1):2176614.
25. Zhao X, Zhang J, Huang Y, Tian Y, Yuan L. Detection and discrimination of disease and insect stress of tea plants using hyperspectral imaging combined with wavelet analysis. *Comput Electron Agric.* 2022;193: 106717.
26. Guo Y, Zhang J, Yin C, Hu X, Zou Y, Xue Z, Wang W. Plant disease identification based on deep learning algorithm in smart farming. *Discret Dyn Nat Soc.* 2020;2020:1–11.
27. Zhang K, Zhang L, Wu Q. Identification of cherry leaf disease infected by *Podosphaera pannosa* via convolutional neural network. *Int J Agric Environ Inf Syst (JAEIS).* 2019;10(2):98–110.
28. Anami BS, Malvade NN, Palaiah S. Classification of yield affecting biotic and abiotic paddy crop stresses using field images. *Inf Process Agric.* 2020;7(2):272–85.
29. Bose P, Dutta S, Goyal V, Bandyopadhyay SK. Leaf diseases detection of medicinal plants based on support vector machine classification algorithm. *J Pharm Res Int.* 2021;33(42A):111–9.
30. Javidan SM, Banakar A, Vakilian KA, Ampatzidis Y. Diagnosis of grape leaf diseases using automatic K-means clustering and machine learning. *Smart Agric Technol.* 2023;3: 100081.
31. Xie Y, Plett D, Evans M, Garrard T, Butt M, Clarke K, Liu H. Hyperspectral imaging detects biological stress of wheat for early diagnosis of crown rot disease. *Comput Electron Agric.* 2024;217: 108571.
32. Kaur S, Pandey S, Goel S. Semi-automatic leaf disease detection and classification system for soybean culture. *IET Image Process.* 2018;12:1038–48.
33. Abisha S, Jayasree T. Application of image processing techniques and artificial neural network for detection of diseases on brinjal leaf. *IETE J Res.* 2022;68(3):2246–58.
34. Jan M, Ahmad H. Image features based intelligent apple disease prediction system: machine learning based apple disease prediction system. *Int J Agric Environ Inf Syst (JAEIS).* 2020;11(3):31–47.

35. Abdulridha J, Ampatzidis Y, Kakarla SC, Roberts P. Detection of target spot and bacterial spot diseases in tomato using UAV-based and benchtop-based hyperspectral imaging techniques. *Precis Agric.* 2020;21:955–78.
36. Rath AK, Meher JK. Disease detection in infected plant leaf by computational method. *Arch Phytopathol Plant Prot.* 2019;52(19–20):1348–58.
37. Singh UP, Chouhan SS, Jain S, Jain S. Multilayer convolution neural network for the classification of Mango leaves infected by Anthracnose disease. *IEEE Access.* 2019;7:43721–9.
38. Deshpande R, Patidar H. Tomato plant leaf disease detection using generative adversarial network and deep convolutional neural network. *Imaging Sci J.* 2022;70(1):1–9.
39. Sujithra J, Ukrit MF. CRUN-based leaf disease segmentation and morphological-based stage identification. *Math Probl Eng.* 2022;2022:2546873.
40. Ferentinos KP. Deep learning models for plant disease detection and diagnosis. *Comput Electron Agric.* 2018;145:311–8.
41. Cruz A, Ampatzidis Y, Pierro R, Materazzi A, Panattoni A, De Bellis L, Luvisi A. Detection of grapevine yellows symptoms in *Vitis vinifera* L. with artificial intelligence. *Comput Electron Agric.* 2019;157:63–76.
42. Hu G, Wu H, Zhang Y, Wan M. A low shot learning method for tea leaf's disease identification. *Comput Electron Agric.* 2019;163: 104852. <https://doi.org/10.1016/j.compag.2019.104852>.
43. Singh G, Yogi KK. Performance evaluation of plant leaf disease detection using deep learning models. *Arch Phytopathol Plant Prot.* 2023;56(3):209–33.
44. Liu X, Min W, Mei S, Wang L, Jiang S. Plant disease recognition: a large-scale benchmark dataset and a visual region and loss reweighting approach. *IEEE Trans Image Process.* 2021;30:2003–15.
45. Jung DH, Kim JD, Kim HY, Lee TS, Kim HS, Park SH. A hyperspectral data 3D convolutional neural network classification model for diagnosis of gray mold disease in strawberry leaves. *Front Plant Sci.* 2022;13: 837020.
46. Gonzalez-Huitron V, León-Borges JA, Rodríguez-Mata AE, Amabilis-Sosa LE, Ramírez-Pereda B, Rodríguez H. Disease detection in tomato leaves via CNN with lightweight architectures implemented in Raspberry Pi 4. *Comput Electron Agric.* 2021;181: 105951.
47. Zhang X, Qiao Y, Meng F, Fan C, Zhang M. Identification of maize leaf diseases using improved deep convolutional neural networks. *IEEE Access.* 2018;6:30370–7.
48. Hassam M, Khan MA, Armghan A, Althubiti SA, Alhaisoni M, Alqahtani A, Kadry S, Kim Y. A single stream modified mobilenet V2 and whale controlled entropy based optimization framework for citrus fruit diseases recognition. *IEEE Access.* 2022;10:91828–39.
49. Zhao Y, Chen Z, Gao X, Song W, Xiong Q, Hu J, Zhang Z. Plant disease detection using generated leaves based on Double-GAN. *IEEE/ACM Trans Comput Biol Bioinf.* 2021;19(3):1817–26.
50. Ahmad I, Hamid M, Yousaf S, Shah ST, Ahmad MO. Optimizing pretrained convolutional neural networks for tomato leaf disease detection. *Complexity.* 2020;2020(1):8812019.
51. Bruno A, Moroni D, Dainelli R, Rocchi L, Morelli S, Ferrari E, Martinelli M. Improving plant disease classification by adaptive minimal ensembling. *Frontiers in Artificial Intelligence.* 2022;5:868926.
52. Ramanadham KL, Savarimuthu N. vCrop: an automated plant disease prediction using deep ensemble framework using real field images. *Sādhanā.* 2022;47(4):268.
53. Bhagat S, Kokare M, Haswani V, Hambarde P, Taori T, Ghante PH, Patil DK. Advancing real-time plant disease detection: a lightweight deep learning approach and novel dataset for pigeon pea crop. *Smart Agric Technol.* 2024;7: 100408.
54. Sharma R, Singh A. Big bang–big crunch-CNN: an optimized approach towards rice crop protection and disease detection. *Arch Phytopathol Plant Protect.* 2022;55(2):143–61.
55. Geetharamani G, Pandian A. Identification of plant leaf diseases using a nine-layer deep convolutional neural network. *Comput Electr Eng.* 2019;76:323–38.
56. Zeng T, Li C, Zhang B, Wang R, Fu W, Wang J, Zhang X. Rubber leaf disease recognition based on improved deep convolutional neural networks with a cross-scale attention mechanism. *Front Plant Sci.* 2022;13: 829479.
57. Chen X, Zhou G, Chen A, Yi J, Zhang W, Hu Y. Identification of tomato leaf diseases based on combination of ABCK-BWTR and B-ARNet. *Comput Electron Agric.* 2020;178: 105730.
58. Yang G, Chen G, He Y, Yan Z, Guo Y, Ding J. Self-supervised collaborative multi-network for fine-grained visual categorization of tomato diseases. *IEEE Access.* 2020;8:211912–23.
59. Kunduracioglu I, Pacal I. Advancements in deep learning for accurate classification of grape leaves and diagnosis of grape diseases. *J Plant Dis Prot.* 2024;131(3):1061–80.
60. Li X, Li S. Transformer help CNN see better: a lightweight hybrid apple disease identification model based on transformers. *Agriculture.* 2022;12(6):884.
61. Balakrishna K, Rao M. Tomato plant leaves disease classification using KNN and PNN. *Int J Comput Vis Image Process (IJCVIP).* 2019;9(1):51–63.
62. Sabrol H, Kumar S. Plant leaf disease detection using adaptive neuro-fuzzy classification. In: *Advances in computer vision: proceedings of the 2019 computer vision conference (CVC), vol. 1*. Springer International Publishing; 2020. pp. 434–443.
63. Bedi P, Gole P. Plant disease detection using hybrid model based on convolutional autoencoder and convolutional neural network. *Artif Intell Agric.* 2021;5:90–101.
64. Wang C, Zhou J, Zhang Y, Wu H, Zhao C, Teng G, Li J. A plant disease recognition method based on fusion of images and graph structure text. *Front Plant Sci.* 2022;12: 731688.
65. Ni C, Wang D, Vinson R, Holmes R, Tao Y. Automatic inspection machine for maize kernels based on deep convolutional neural networks. *Biosys Eng.* 2019;178:131–44.
66. Kusumo BS, Heryana A, Krisnandi D, Yuwana S, Zilvan V, Pardede HF. Deep convolutional neural networks-based plants diseases detection using hybrid features. *Comput Eng Appl J.* 2020;9(3):215–25.
67. Sujatha R, Chatterjee JM, Jhanjhi NZ, Brohi SN. Performance of deep learning vs machine learning in plant leaf disease detection. *Microprocess Microsyst.* 2021;80: 103615.
68. Girmaw DW, Muluneh TW. Deep convolutional neural network model for classifying common bean leaf diseases. *Discover Artif Intell.* 2019;4(1):96.
69. Tetila EC, Machado BB, Menezes GK, Oliveira ADS, Alvarez M, Amorim WP, Pistori H. Automatic recognition of soybean leaf diseases using UAV images and deep convolutional neural networks. *IEEE geoscience and remote sensing letters.* 2019;17(5):903–907.

70. Liu B, Tan C, Li S, He J, Wang H. A data augmentation method based on generative adversarial networks for grape leaf disease identification. *IEEE Access*. 2020;8:102188–98.
71. Wu Q, Chen Y, Meng J. DCGAN-based data augmentation for tomato leaf disease identification. *IEEE Access*. 2020;8:98716–28.
72. Zhong Y, Zhao M. Research on deep learning in apple leaf disease recognition. *Comput Electron Agric*. 2020;168: 105146.
73. Luo Y, Sun J, Shen J, Wu X, Wang L, Zhu W. Apple leaf disease recognition and sub-class categorization based on improved multi-scale feature fusion network. *IEEE Access*. 2021;9:95517–27.
74. Chowdhury ME et al. Automatic and reliable leaf disease detection using deep learning techniques. *AgriEngineering*. 2021;3(2):294–312.
75. Bansal P, Kumar R, Kumar S. Disease detection in apple leaves using deep convolutional neural network. *Agriculture*. 2021;11(7):617.
76. Bakr M, Abdel-Gaber S, Nasr M, Hazman M. Tomato disease detection model based on densenet and transfer learning. *Appl Comput Sci*. 2022;18(2).
77. Nirmal MD, Jadhav PP, Pawar S. Pomegranate leaf disease detection using supervised and unsupervised algorithm techniques. *Cybern Syst*. 2023;1–12.
78. Mahum R et al. A novel framework for potato leaf disease detection using an efficient deep learning model. *Human Ecol Risk Assess Int J*. 2023;29(2):303–326.
79. Yang D, Wang F, Hu Y, Lan Y, Deng X. Citrus huanglongbing detection based on multi-modal feature fusion learning. *Front Plant Sci*. 2021;12: 809506.
80. Elaraby A, Hamdy W, Alanazi S. Classification of citrus diseases using optimization deep learning approach. *Comput Intell Neurosci*. 2022;2022(1):9153207.
81. Barbedo JGA. Plant disease identification from individual lesions and spots using deep learning. *Biosys Eng*. 2019;180:96–107.
82. Hassan SM, Maji AK, Jasiński M, Leonowicz Z, Jasińska E. Identification of plant-leaf diseases using CNN and transfer-learning approach. *Electronics*. 2021;10(12):1388.
83. Lv M, Zhou G, He M, Chen A, Zhang W, Hu Y. maize leaf disease identification based on feature enhancement and DMS-Robust Alexnet. *IEEE Access*. 2020;8:57952–66. <https://doi.org/10.1109/ACCESS.2020.2982443>.
84. Wu Q, Ji M, Deng Z. Automatic detection and severity assessment of pepper bacterial spot disease via multimodels based on convolutional neural networks. *Int J Agric Environ Inf Syst (IAEIS)*. 2020;11(2):29–43.
85. Janarthan S, Thuseethan S, Rajasegarar S, Lyu Q, Zheng Y, Yearwood J. Deep metric learning based citrus disease classification with sparse data. *IEEE Access*. 2020;8:162588–600.
86. Chen J, Zhang D, Nanehkaran YA, Li D. Detection of rice plant diseases based on deep transfer learning. *J Sci Food Agric*. 2020;100(7):3246–56.
87. Chen J, Chen J, Zhang D, Sun Y, Nanehkaran YA. Using deep transfer learning for image-based plant disease identification. *Comput Electron Agric*. 2020;173: 105393.
88. Chen J, Chen W, Zeb A, Yang S, Zhang D. Lightweight inception networks for the recognition and detection of rice plant diseases. *IEEE Sens J*. 2022;22(14):14628–38.
89. Liu D, Yang H, Gong Y, Chen Q. A recognition method of crop diseases and insect pests based on transfer learning and convolution neural network. *Math Probl Eng*. 2022;2022(1):1470541.
90. Wang H, Pan X, Zhu Y, Li S, Zhu R. Maize leaf disease recognition based on TC-MRSN model in sustainable agriculture. *Comput Electron Agric*. 2024;221: 108915.
91. Chillakuru P, Divya D, Ananthajothi K. Enhanced segmentation with optimized nine-layered CNN-based classification of leaf diseases: an automatic approach for plant disease diagnosis. *Cybern Syst*. 1–36.
92. Durairaj V, Surianarayanan C. Disease detection in plant leaves using segmentation and autoencoder techniques. *Malaya J*. 2020.
93. Syarif M, Setiawan W. Convolutional neural network for maize leaf disease image classification. *Telkomnika (Telecommun Comput Electron Control)*. 2020;18(3):1376–81.
94. Kainat J, Sajid Ullah S, Alharithi FS, Alroobaea R, Hussain S, Nazir S. Blended features classification of leaf-based cucumber disease using image processing techniques. *Complexity*. 2021;2021(1):9736179.
95. Singh AK, Sreenivasu SVN, Mahalaxmi USBK, Sharma H, Patil DD, Asenso E. Hybrid feature-based disease detection in plant leaf using convolutional neural network, bayesian optimized SVM, and random forest classifier. *J Food Qual*. 2022;2022:1–16.
96. Pantazi XE, Moshou D, Tamouridou AA. Automated leaf disease detection in different crop species through image features analysis and one class classifiers. *Comput Electron Agric*. 2019;156:96–104.
97. Khan MA, Lali MIU, Sharif M, Javed K, Aurangzeb K, Haider SI, Akram T. An optimized method for segmentation and classification of apple diseases based on strong correlation and genetic algorithm based feature selection. *IEEE Access*. 2019;7:46261–46277.
98. Nagasubramanian G, Sakthivel RK, Patan R, Sankayya M, Daneshmand M, Gandomi AH. Ensemble classification and IoT-based pattern recognition for crop disease monitoring system. *IEEE Internet Things J*. 2021;8(16):12847–54.
99. Zhang S, Wang H, Huang W, You Z. Plant diseased leaf segmentation and recognition by fusion of superpixel, K-means and PHOG. *Optik*. 2018;157:866–72.
100. Oo YM, Htun NC. Plant leaf disease detection and classification using image processing. *Int J Res Eng*. 2018;5(9):516–23.
101. Hussein MA, Abbas AH. Plant leaf disease detection using support vector machine. *Al-Mustansiriyah J Sci*. 2019;30(1):105–10.
102. Zamani AS, Anand L, Rane KP, Prabhu P, Buttar AM, Pallathadka H, Dugbakie BN. [Retracted] Performance of machine learning and image processing in plant leaf disease detection. *J Food Qual*. 2022;2022(1):1598796.
103. Sun Y, Jiang Z, Zhang L, Dong W, Rao Y. SLIC_SVM based leaf diseases saliency map extraction of tea plant. *Comput Electron Agric*. 2019;157:102–9.
104. Mukhopadhyay S, Paul M, Pal R, De D. Tea leaf disease detection using multi-objective image segmentation. *Multimed Tools Appl*. 2021;80:753–71.
105. Yashodharan S. Neural network for papaya leaf disease detection. *Acta Gr Znanstveni Čas Tiskarstvo Gr Komun*. 2019;30(3):11–24.

106. Ma J, Du K, Zheng F, Zhang L, Gong Z, Sun Z. A recognition method for cucumber diseases using leaf symptom images based on deep convolutional neural network. *Comput Electron Agric.* 2018;154:18–24.
107. Yu HJ, Son CH. Apple leaf disease identification through region-of-interest-aware deep convolutional neural network. arXiv preprint [arXiv:1903.10356](https://arxiv.org/abs/1903.10356). 2019.
108. Picon A, Seitz M, Alvarez-Gila A, Mohnke P, Ortiz-Barredo A, Echazarra J. Crop conditional convolutional neural networks for massive multi-crop plant disease classification over cell phone acquired images taken on real field conditions. *Comput Electron Agric.* 2019;167: 105093.
109. Loyani LK, Bradshaw K, Machuve D. Segmentation of tuta absoluta's damage on tomato plants: a computer vision approach. *Appl Artif Intell.* 2021;35(14):1107–27.
110. Wu Y, Feng X, Chen G. Plant leaf diseases fine-grained categorization using convolutional neural networks. *IEEE Access.* 2022;10:41087–96.
111. Liu Y, Gao G, Zhang Z. Crop disease recognition based on modified light-weight CNN with attention mechanism. *IEEE Access.* 2022;10:112066–75.
112. Abouelmagd LM, Shams MY, Marie HS, Hassanien AE. An optimized capsule neural networks for tomato leaf disease classification. *EURASIP J Image Video Proc.* 2024;2024(1):2.
113. Qian X, Zhang C, Chen L, Li K. Deep learning-based identification of maize leaf diseases is improved by an attention mechanism: self-attention. *Front Plant Sci.* 2022;13: 864486.
114. Rao A, Kulkarni SB. A hybrid approach for plant leaf disease detection and classification using digital image processing methods. *Int J Electric Eng Educ.* 2020;0020720920953126.
115. Davida HE, Ramalakshmib K, Venkatesanc R, Hemalathad G. Tomato leaf disease detection using hybrid CNN-RNN model. *Adv Parallel Comput.* 2021;38:593–7.
116. Sowmya BJ, Shetty C, Seema S, Srinivasa KG. An image processing and machine learning approach for early detection of diseased leaves. *Int J Cyber-Phys Syst (IJCPS).* 2019;1(2):56–73.
117. Javidan SM, Banakar A, Rahnama K, Vakilian KA, Ampatzidis Y. Feature engineering to identify plant diseases using image processing and artificial intelligence: a comprehensive review. *Smart Agric Technol.* 2024;100480.
118. Khare O, Mane S, Kulkarni H, Barve N. LeafNST: an improved data augmentation method for classification of plant disease using object-based neural style transfer. *Discover Artif Intell.* 2024;4(1):50.
119. Pacal I, Kunduracioglu I, Alma MH, Deveci M, Kadry S, Nedoma J, Martinek R. A systematic review of deep learning techniques for plant diseases. *Artif Intell Rev.* 2024;57(11), 304.
120. Chouhan SS, Singh UP, Jain S. Performance evaluation of different deep learning models used for the purpose of healthy and diseased leaves classification of Cherimoya (*Annona Cherimola*) plant. *Neural Comput Appl.* 2025;37(6):4531–44.

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